Experiment 5

AIM: Deploy a simple application using Nginx / Apache.

Step 1: Create Simple Node.js App
1. Create a Project

```
C:\Yash EX5>mkdir nginx-demo

C:\Yash EX5>cd nginx-demo

C:\Yash EX5\nginx-demo>npm init -y
(node:8012) ExperimentalWarning: CommonJS module C:\Program
Files\nodejs\node_modules\npm\node_modules\debug\src\node.js is loading ES
Module C:\Program
Files\nodejs\node_modules\npm\node_modules\supports-color\index.js using
require().
Support for loading ES Module in require() is an experimental feature and
might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
Wrote to C:\Yash EX5\nginx-demo\package.json:

{
  "name": "nginx-demo",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

C:\Yash EX5\nginx-demo>npm install express
(node:3168) ExperimentalWarning: CommonJS module C:\Program
Files\nodejs\node_modules\npm\node_modules\debug\src\node.js is loading ES
Module C:\Program
Files\nodejs\node_modules\npm\node_modules\supports-color\index.js using
require().
Support for loading ES Module in require() is an experimental feature and
might change at any time
```

```
(Use `node --trace-warnings ...` to show where the warning was created)

added 65 packages, and audited 66 packages in 1s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

2. Create index.js

C:\Yash EX5\nginx-demo>notepad index.js

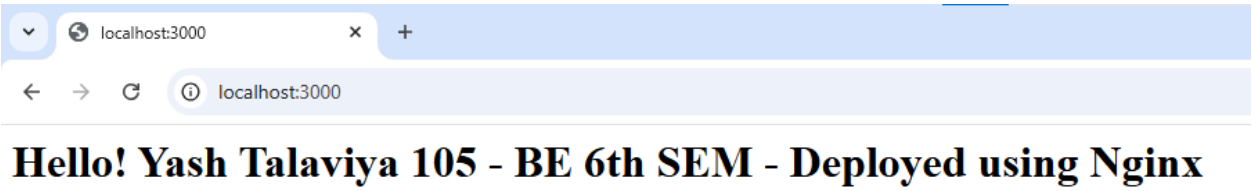
Add this code in index.js file:

```
const express = require("express");
const app = express();
app.get("/", (req, res) => {
res.send("<h1>Hello! Yash Talaviya 105 - BE 6th SEM - Deployed using Nginx
</h1>");
});
app.get("/about", (req, res) => {
res.send("<h2>About Page - Nginx Reverse Proxy Demo </h2>");
});
app.listen(3000, () => {
console.log("App running on http://localhost:3000");
});
```

3. Run the application

```
C:\Yash EX5\nginx-demo>node index.js
App running on http://localhost:3000
```

Output at http://localhost:3000 is:



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In New CMD Terminal:

Step 2: Setup Nginx on Windows

1. Download Nginx

C:\nginx [verify it]

2. Configure nginx.conf

C:\>cd nginx

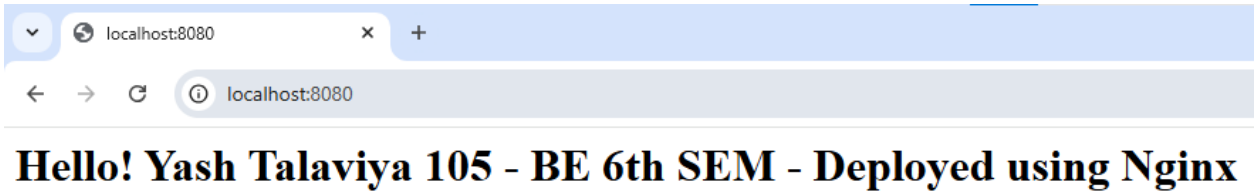
```
C:\nginx>cd conf

C:\nginx\conf>notepad nginx.conf
```

Modify code of **nginx.conf** :

```
worker_processes 1;

events {
    worker_connections 1024;
}

http {

    server {
        listen 8080;
        server_name localhost;

        location / {
            proxy_pass http://localhost:3000;
        }

    }

}

C:\nginx\conf>cd ..
```

3. Start Nginx

```
C:\nginx>nginx -t
nginx: the configuration file C:\nginx/conf/nginx.conf syntax is ok
nginx: configuration file C:\nginx/conf/nginx.conf test is successful

C:\nginx>nginx
```



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